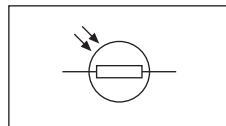
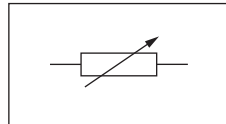
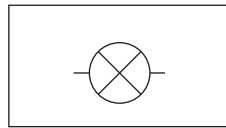
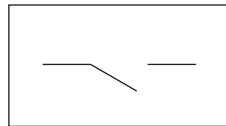


Answer **all** questions.

1. Students are investigating the properties of series and parallel circuits.

- (a) **Draw a straight line** linking the circuit symbol on the left with the name of the component on the right. [3]



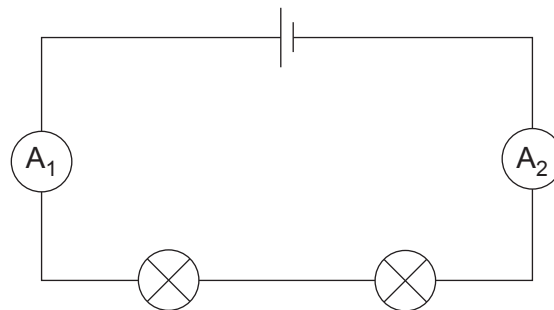
lamp

switch

variable
resistor

LDR

- (b) The students set up the following circuit and measure the current in different places.



- (i) Underline words or phrases from the brackets to correctly complete the following sentences. [2]

The circuit is a (**double** / **series** / **parallel**) circuit.

The current reading on ammeter A_2 is (**less than** / **the same as** / **more than**) the reading on ammeter A_1 .



- (ii) The battery voltage in the circuit is 3V and the current through the lamps is 0.25A.

I. Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the circuit.

[2]

power = W

II. Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the circuit.

[2]

resistance =

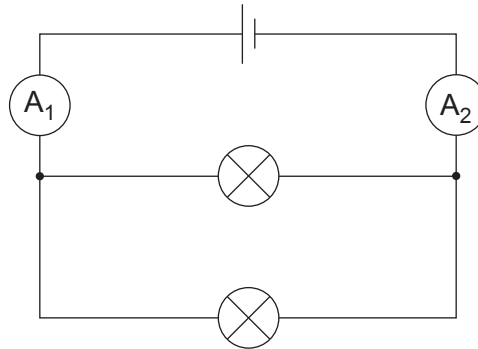
- III. **Circle** the correct unit for resistance.

[1]

J Ω kg



(c) The students now use the **same** components to make the following circuit.



The students notice that the lamps are **brighter**.

Select words or phrases from the box to correctly complete the sentences below.
You may use each word once, more than once or not at all.

[2]

increases	decreases	stays the same
-----------	-----------	----------------

The lamps are brighter because the resistance of the circuit

so the current through each lamp

3420U101
05

12

